SMART INDIA HACKATHON 2025

- **Problem Statement ID – SIH25137**
- **Problem Statement Title- Tourist Tracker With Blockchain**
- **Theme- Travel & Tourism**
- **PS Category- Software**
- **Team ID- 91848**
- **Team Name – Big 6**

Big 6

Tourist Tracking with Block Chain

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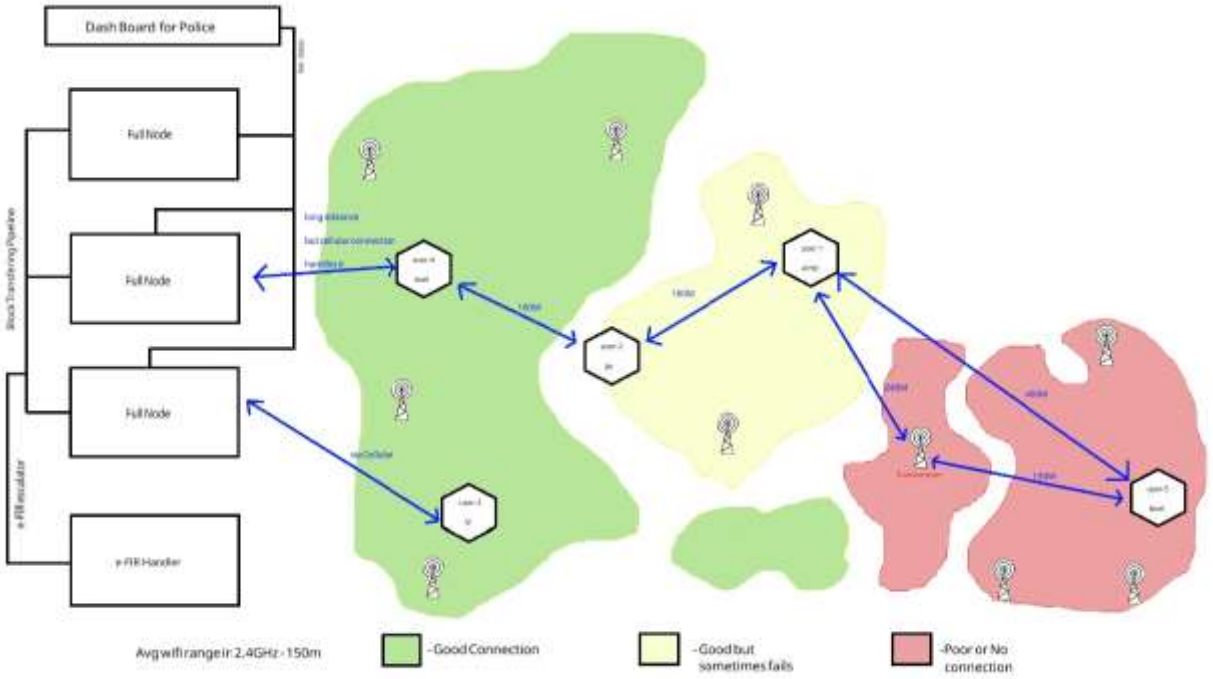
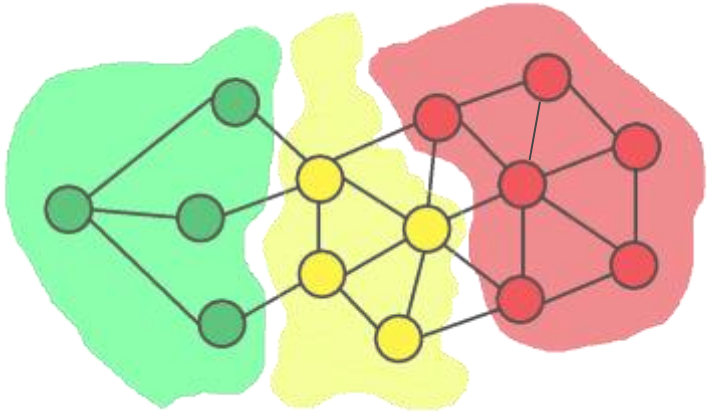
❖ Proposed Solution

- Updates location to server
- BLE based p2p communication implemented
- This p2p activates when internet fails.
- So even without internet, user connects to server.
- Cache location to help when sky is not clear
- Ai based request priority manager
- transceivers placed around
- With offline (map + incident response) system



Project - ARCHITECTURE



Transreciever Placement Strategy

BLE- Bluetooth Low Energy
P2p- Peer to Peer

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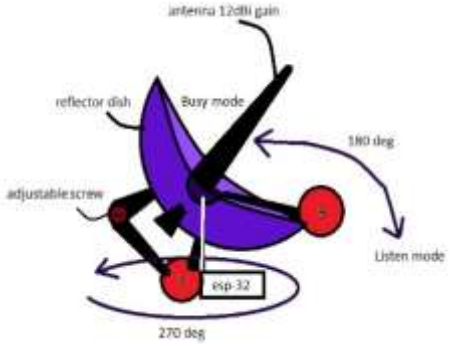
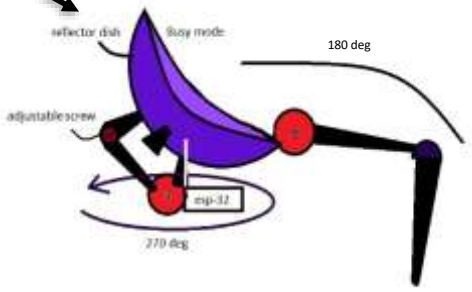
TECHNICAL APPROACH

Flutter App
For android/iOS support

Transceiver
(busy mode)
Range > 0.5Km





Transceiver
(active mode)
Range - 200m

Full node block server

Full node block server

Full node block server

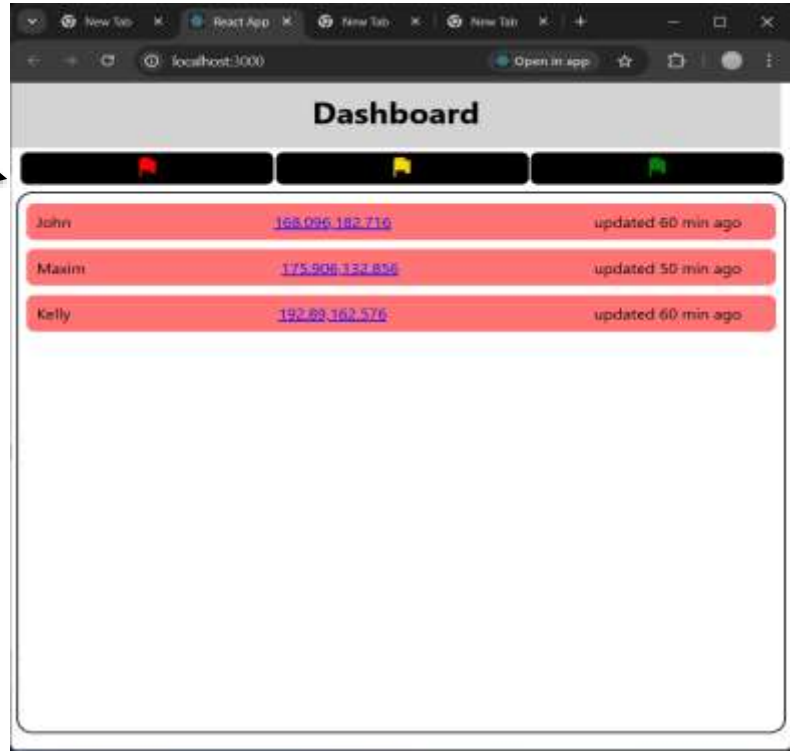
Integrity pipe line

Websocket for realtime updates

E-FIR server

Python servers with **blockchain** implementation

E-FIR server is **separated** from normal node servers **for security**



React.js with **websocket** for **realtime** update

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FEASIBILITY AND VIABILITY

- Can run on any smart phones with necessary permissions.
- Its more cheap and costs only 1800 INR even with high-end components
- Using our placement strategy we can easily cover even bigger range with small number of transceivers
- Likely 12 can cover 1Km radius so just with 100's we can cover 50+Kms radius

Problems:

- Battery life of low-end or old devices.
- BLE range is initially low

Solution :

- Fallback strategy used to drop data-rate, using long scan window. this trades range for lower data rates
- Used timestamp based open window scan that runs for 5s every 10s.
- This timestamp helps in open window syncing.
- This reduces power consumption by half.

Incident response Map

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IMPACT AND BENEFITS

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- This gives confidence to the user to explore without fear.
- The offline + app’s alert system works offline.
- Alerts were triggered when user approach danger zone.
- This helps in reducing the missing cases in tourist areas.
- This makes tourist spots safer.

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RESEARCH AND REFERENCES

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Existing Projects :

- Wearable IoT tracker
- No peer to peer connection when no internet
- No offline map assist
- No blockchain based servers

Our Project:

- It’s a user-friendly mobile app easy to use
- Even with no internet it try to connect to peers for help
- Offline map support for user assist
- Alert system works offline when user enters danger zones
- SOS requests are processed with higher priority for faster response
- Allows custom message with SOS for better helping strategy

Research Paper link : https://fuchsia-gerianne-5.tiiny.site